

LAB #1 – Basic MATLAB Application

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Purpose:

- Learn MATLAB Application and do simple problems.

Before the lab:

- Read the course slides and chapter 1 of the textbook.
- This is an individual lab.
- From this week onwards, all labs will be graded. Please submit your lab report using the provided lab report template by the posted deadline.

Learning outcomes:

Familiarize with mathematical operations using MATLAB.

Use programs with simple scalar variables to solve simple engineering mathematical and engineering problems.

Procedures

Using MATLAB as a calculator:

The most straightforward way to use MATLAB is to use it as a simple calculator. However, you need to pay attention to the order of precedence since it impacts the calculation result in many unexpected ways if misused.

Precedence	Mathematical Operation
First	Parentheses. For nested parentheses, the innermost are executed first.
Second	Exponentiation.
Third	Multiplication, division (equal precedence).
Fourth	Addition and subtraction.

Table 1. The Order of Precedence.

Based on this table, the following tutorial from the textbook illustrate how MATLAB order of precedence works.

During the lab:

1. Open the MATLAB software.
2. In the command window try to enter a number like 190 and see what happens.
3. In the command window try to enter simple expressions like $8*7$ or $100+12$ and see what happens.
4. In the command window enter the following: `num1 = -1`. It stores the number -1 into a container named 'num1'.
5. In the command window enter the following: `num2 = 90`. It stores the value 90 into a container named 'num2'.
6. Now try to enter `num1 / num2` in the command window and see what happens.
7. Try 10 more combinations of numbers, containers and operators and see what happens, write the result.
8. With MATLAB, do the simple problems numbers 1 to 10 in your textbook (section 1.10).

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- Please keep the codes and results in the .m file, please include the codes and results in your report.
9. In your textbook, look at the more advanced problems (section 1.10), select number 21 and 25 and do them with MATLAB. Keep the result and add them to your report.
 10. Close MATLAB.
 11. **Tracing Exercises:** Manually evaluate what will be the values of **x**, **y**, and **z** after the following MATLAB formatted instructions are executed (**Do these by hand: Do not use MATLAB in this practice!**). Must show all your calculation procedures. Record your results in the lab report.

```
aa = 130 + 4;  
bb = 5 / 10 + 2 - 4;  
cc = 5 % 2 * 11;  
x = aa - (bb * cc);  
m1 = 2 ^ 4 - 10;  
k1 = 3 ^ (5 - 11);  
f1 = 87 - 2 / 20;  
y = m1 * k1 - 10 - f1 / 3 ;  
dd = 4.5 * 2.3 - 10.3;  
ee = 5.3 + 6.7 - 0.9;  
j = abs (ee) + round (ee)*9 - fix (ee)*10;  
z = ceil (ee) - sqrt (dd*dd) * floor (ee) - j;
```

Lab Report Submission Requirement:

1. Submit Lab #1 based on the given templates on the Blackboard. Please post all the screenshots of the inputs and running outputs of each step on the lab manual.